Basic Science Department

BAS 141
Engineering Chemistry

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Chapter 3: Thermochemistry

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1.1. Introduction

- Thermochemistry is a branch of thermodynamics that focuses on the absorption or release of heat energy that accompanies chemical reactions
- Study of the heat change that takes place during reactions



Why do chemical reactions occur spontaneously?

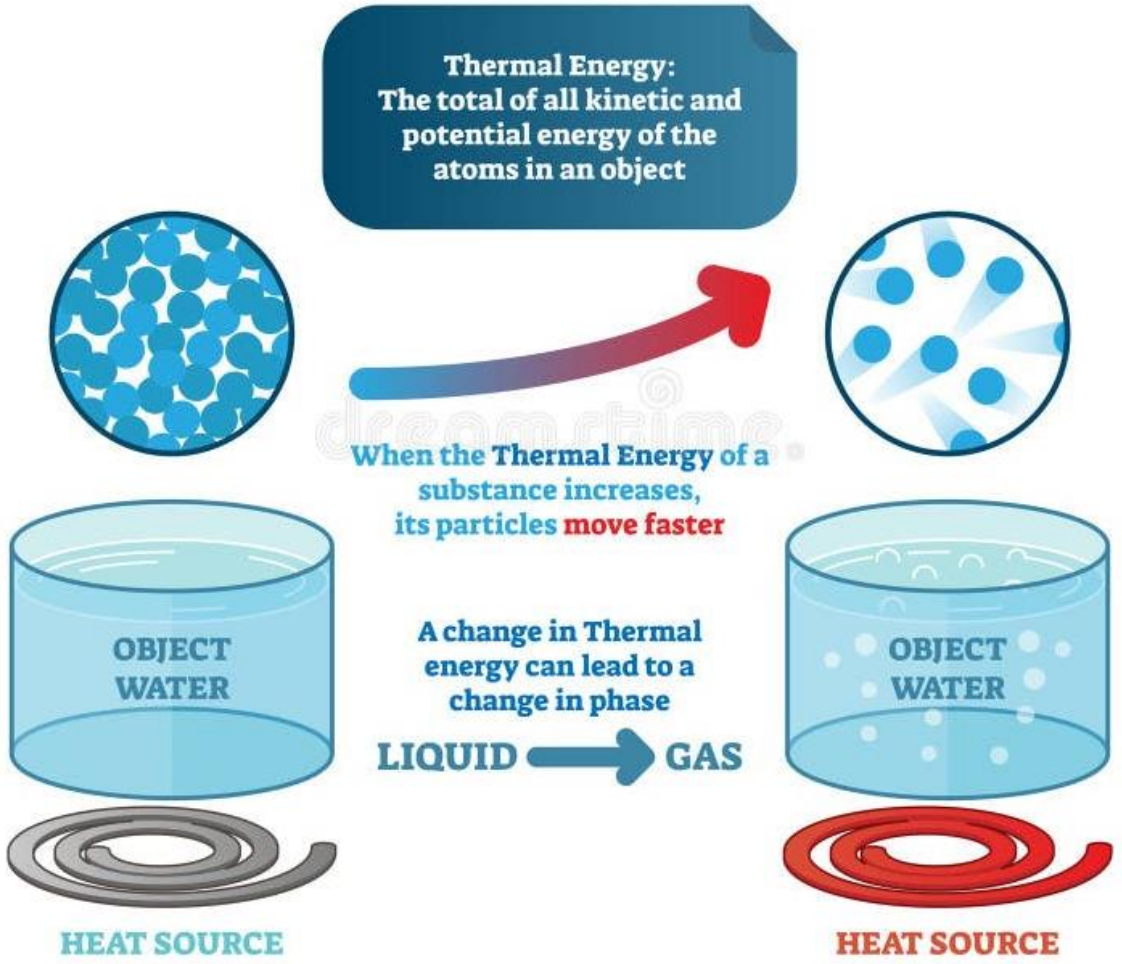
What is “stability” and what does it mean to say that one substance is more stable than another?



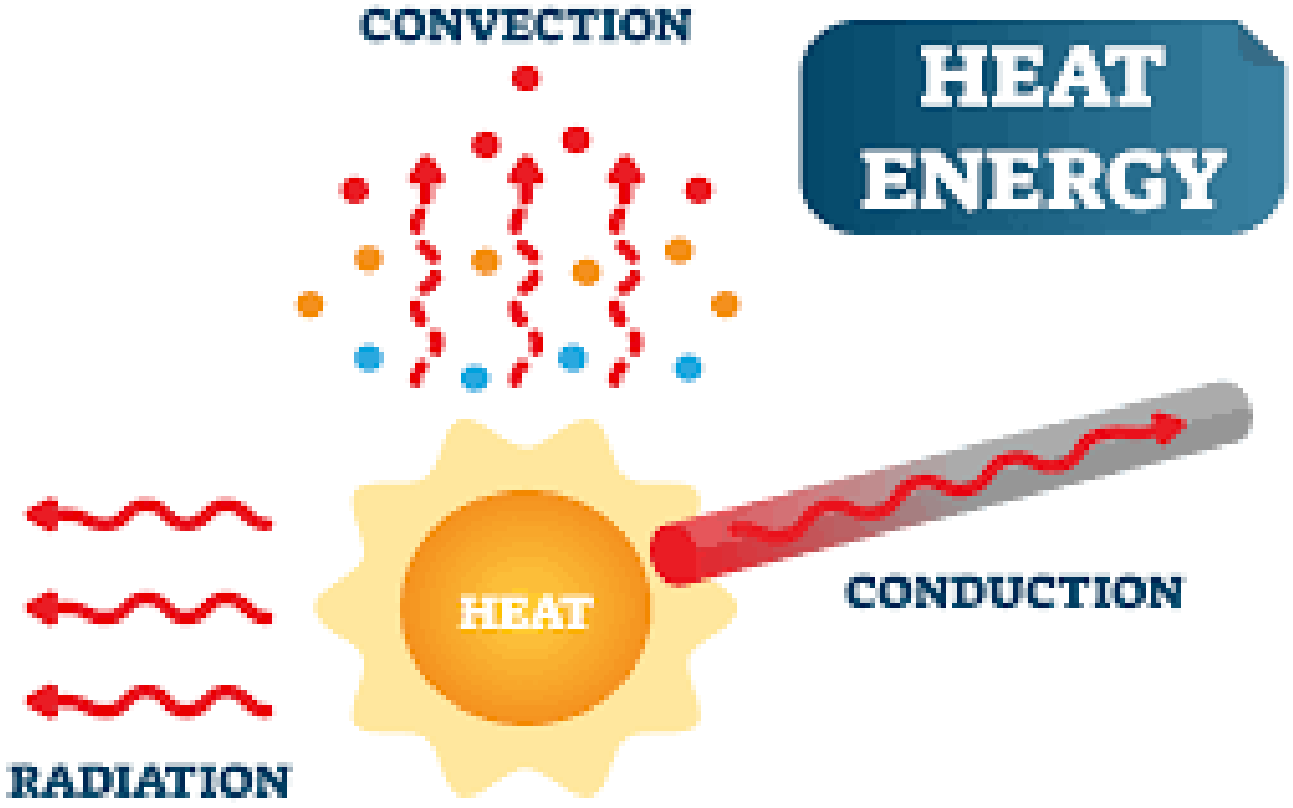
The key factor in determining the stability of a substance is energy.

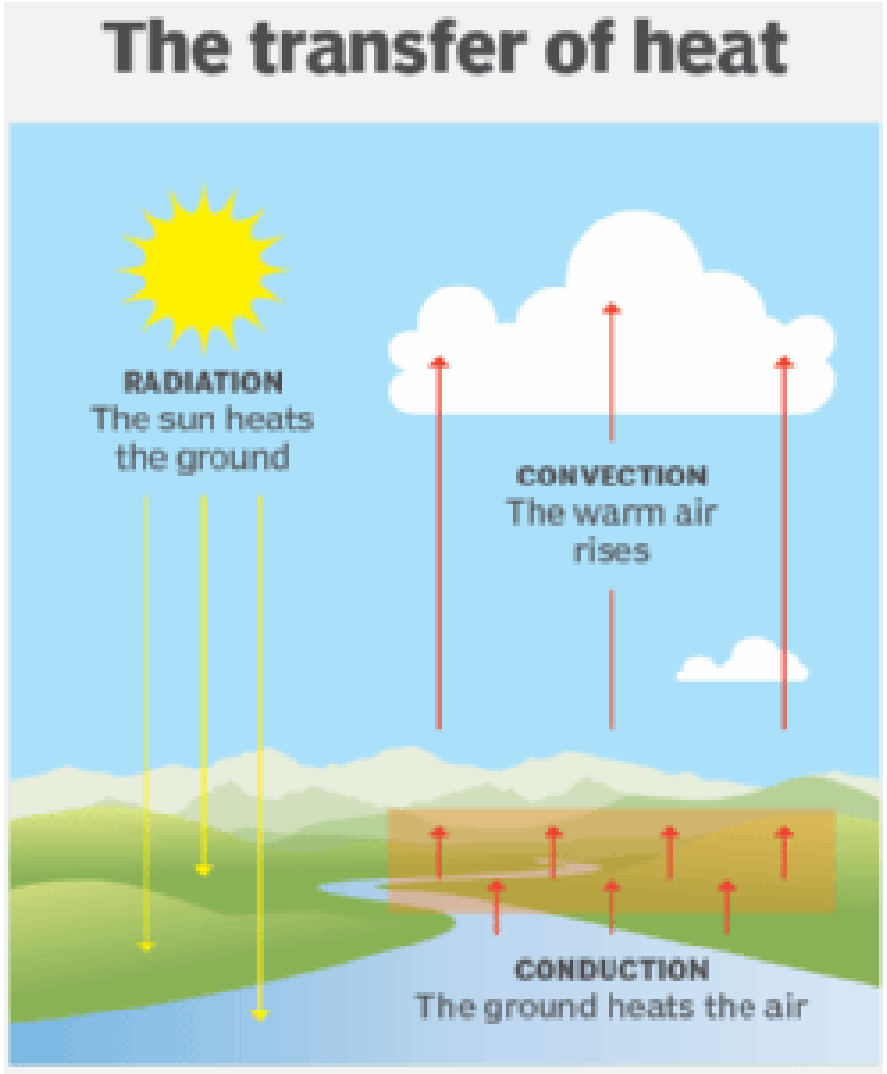
Thermal energy is just the kinetic energy of molecular motion, which we measure by finding the temperature of an object.

- An object has a **low temperature**, its atoms or molecules **move slowly**.
- An object has a **high temperature,** its atoms or molecules **move rapidly** and collide forcefully.

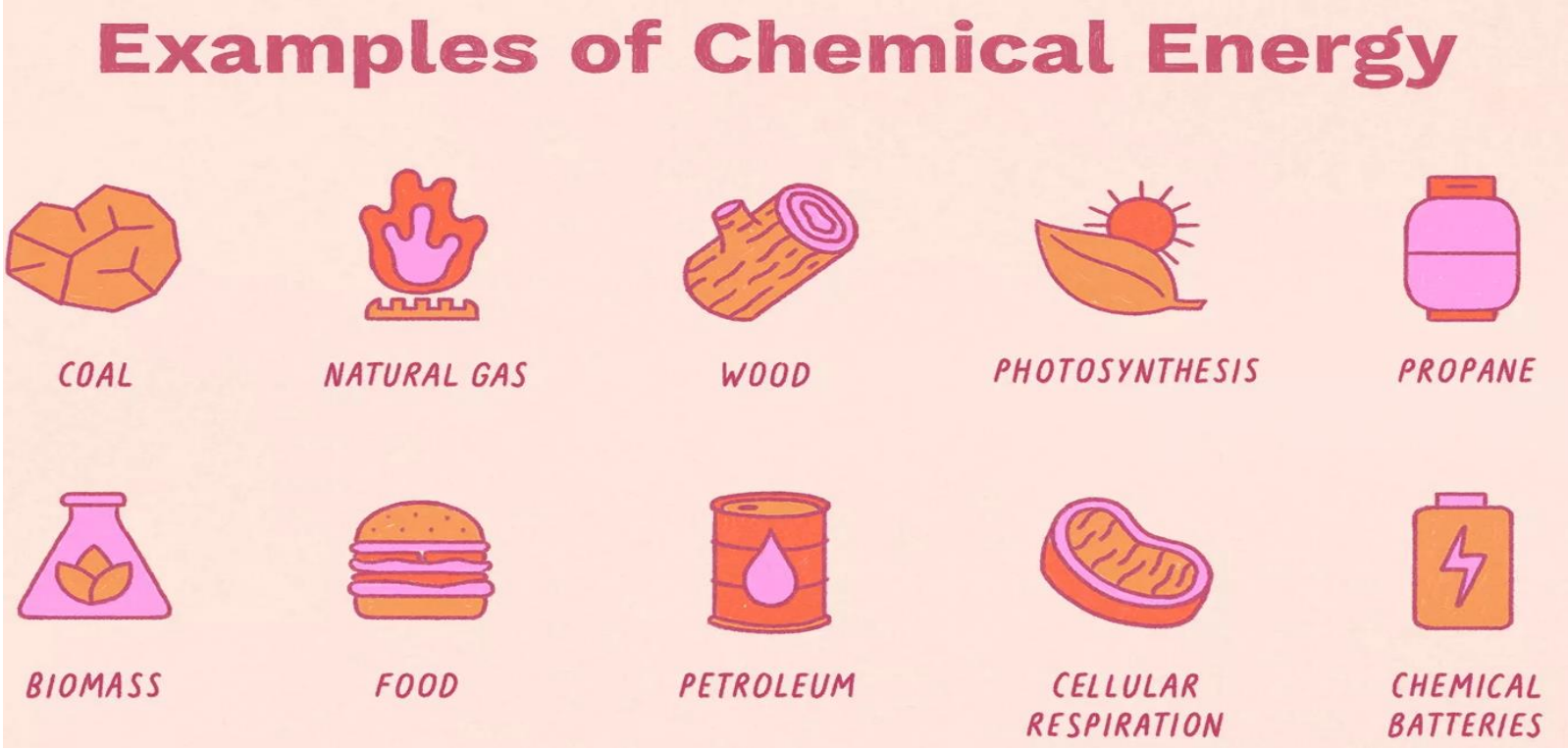


Heat is the amount of thermal energy transferred from one object to another because of a temperature difference.





Chemical energy is a kind of potential energy in which chemical bonds act as the storage medium.



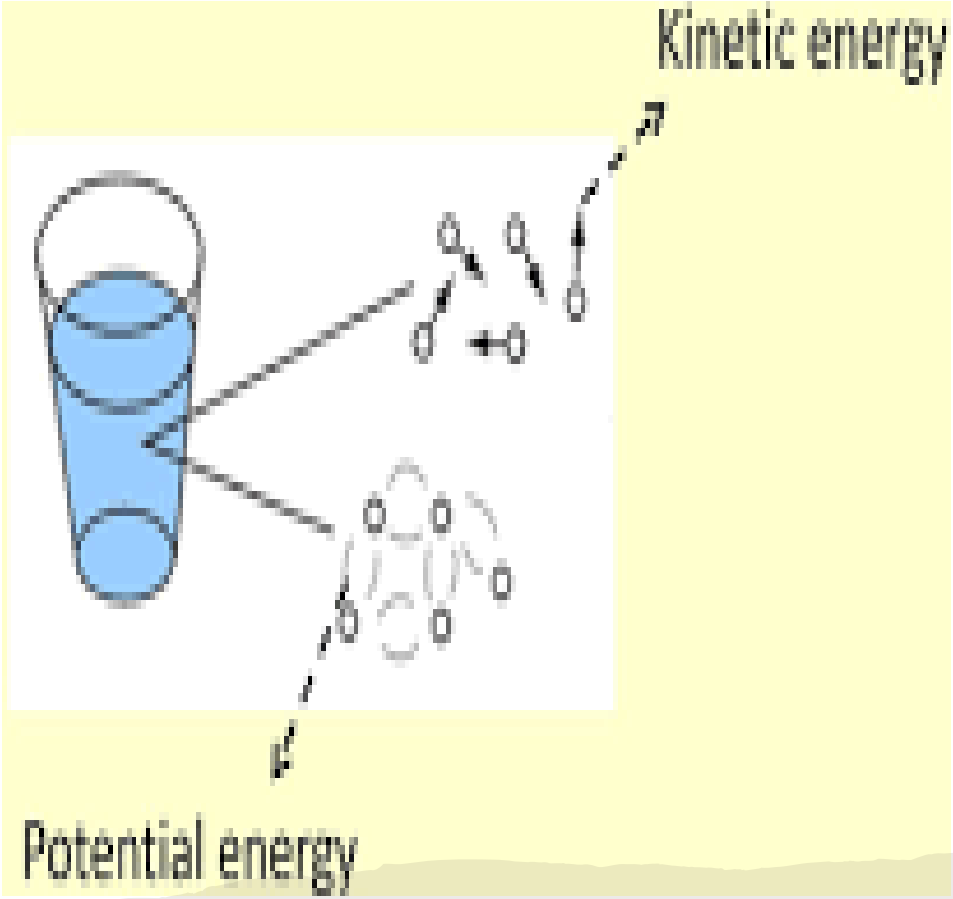
Chemicals can release their potential energy in the form of **heat** or **light** when they undergo reactions and form more stable products.

1.2. Internal energy

The system's Internal Energy (E) is the sum of all the kinetic and potential energies for every system molecule or ion.

- Internal energy is microscopic energy; it deals with energy on molecular levels

- We can measure any change in internal energy (ΔE)



The energy change is the difference between final and initial states.

$$\Delta E = E_{final} - E_{initial}$$

Surroundings

The **system** is the mixture of reactants, and the *surroundings* are the flask, the solvent, the room, and the rest of the universe.

System



Energy flowing out of the system to the surroundings has a negative sign because $E_{final} < E_{initial}$.

Energy flowing into the system from the surroundings has a positive sign because $E_{final} > E_{initial}$.

The internal energy of a system depends on many things:

1. Chemical identity
2. Sample size
3. Temperature
4. Pressure
5. Physical state (gas, liquid, or solid)

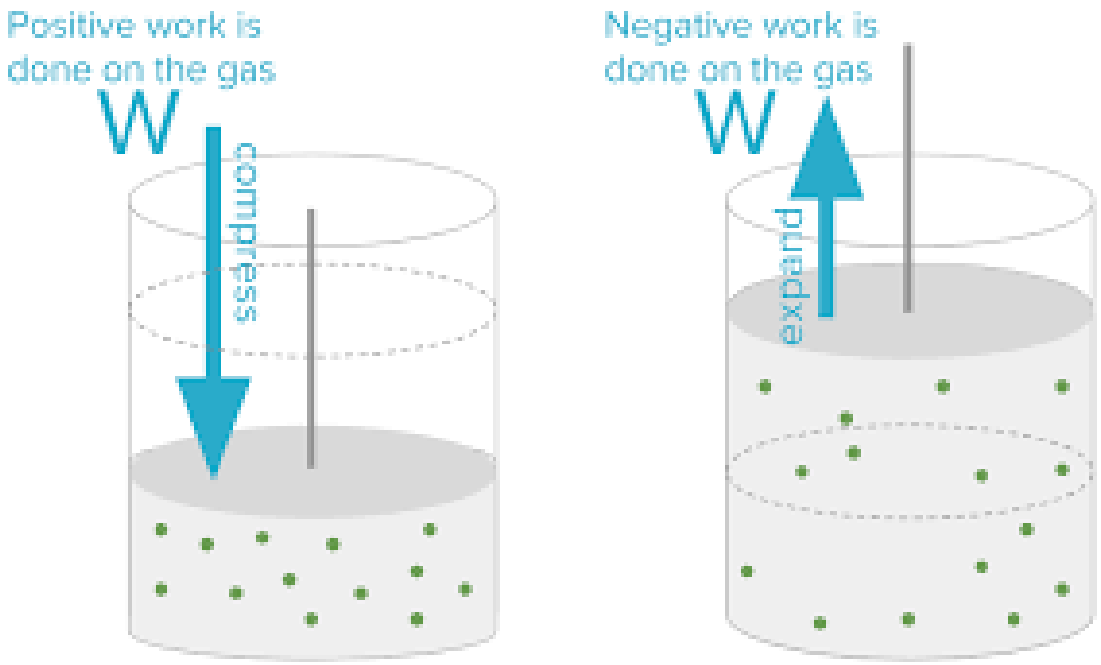
Notes:

- Internal energy does not depend on the system's history.
- Internal energy is said to be a **state function**

1.3. Expansion work

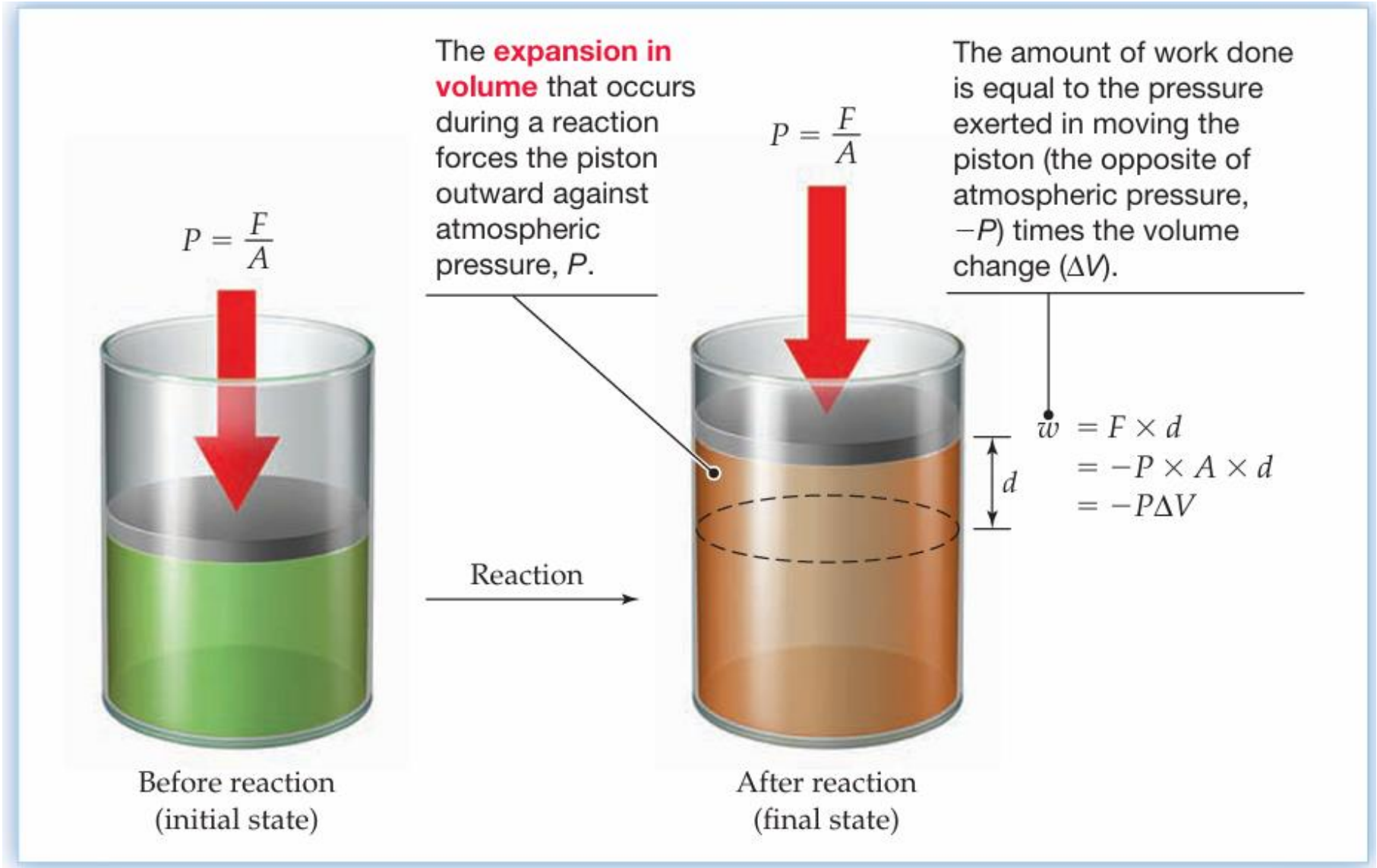
$$\text{Work} = \text{Force} \times \text{Distance}$$

$$w = F \times d$$



The most common type of work in chemical systems is **expansion work (also called pressure–volume, or PV, work)** done as the result of a volume change in the system.

$$\underbrace{C_3H_8(g) + 5\,O_2(g)}_{\text{6 mol of gas}} \longrightarrow \underbrace{3\,CO_2(g) + 4\,H_2O(g)}_{\text{7 mol of gas}}$$



A negative value → $w = -P\Delta V$ ← A positive value

Work done during expansion

Solved example

Calculate the work done during a reaction in which the volume expands from 12.0 L to 14.5 L against an external pressure of 5.0 atm.

Solution

$$\Delta V = (14.5 - 12.0)\,\text{L} = 2.5\,\text{L}$$

$$w = -(5.0\,\text{atm})(2.5\,\text{L}) = -12.5\,\text{L}\cdot\text{atm} \quad = -1.3\,\text{kJ}$$

1.4. Energy and enthalpy

The total change in internal Δ E

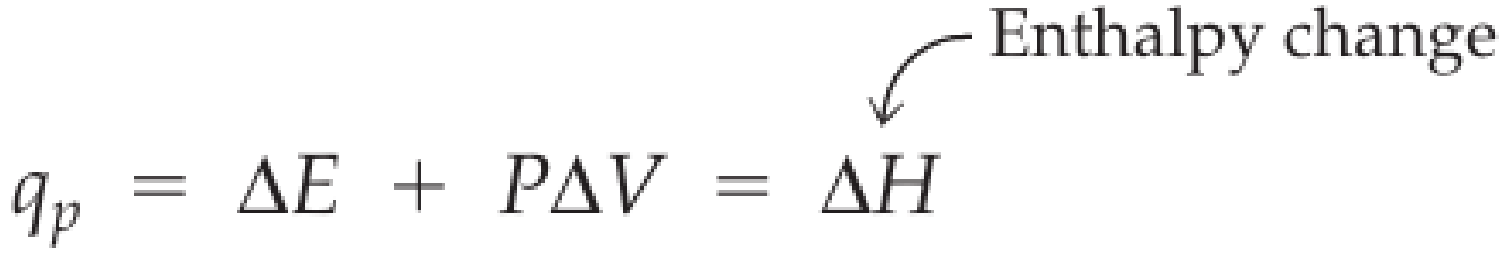
$$\Delta E = q + w = q - P\Delta V$$

- **q** represents transferred heat and has a **positive sign** if the system **gains heat** and a **negative sign** if the system **loses heat.**

heat of the reaction, or the enthalpy change (ΔH)

$$q_p = \Delta E + P\Delta V = \Delta H$$

Enthalpy change

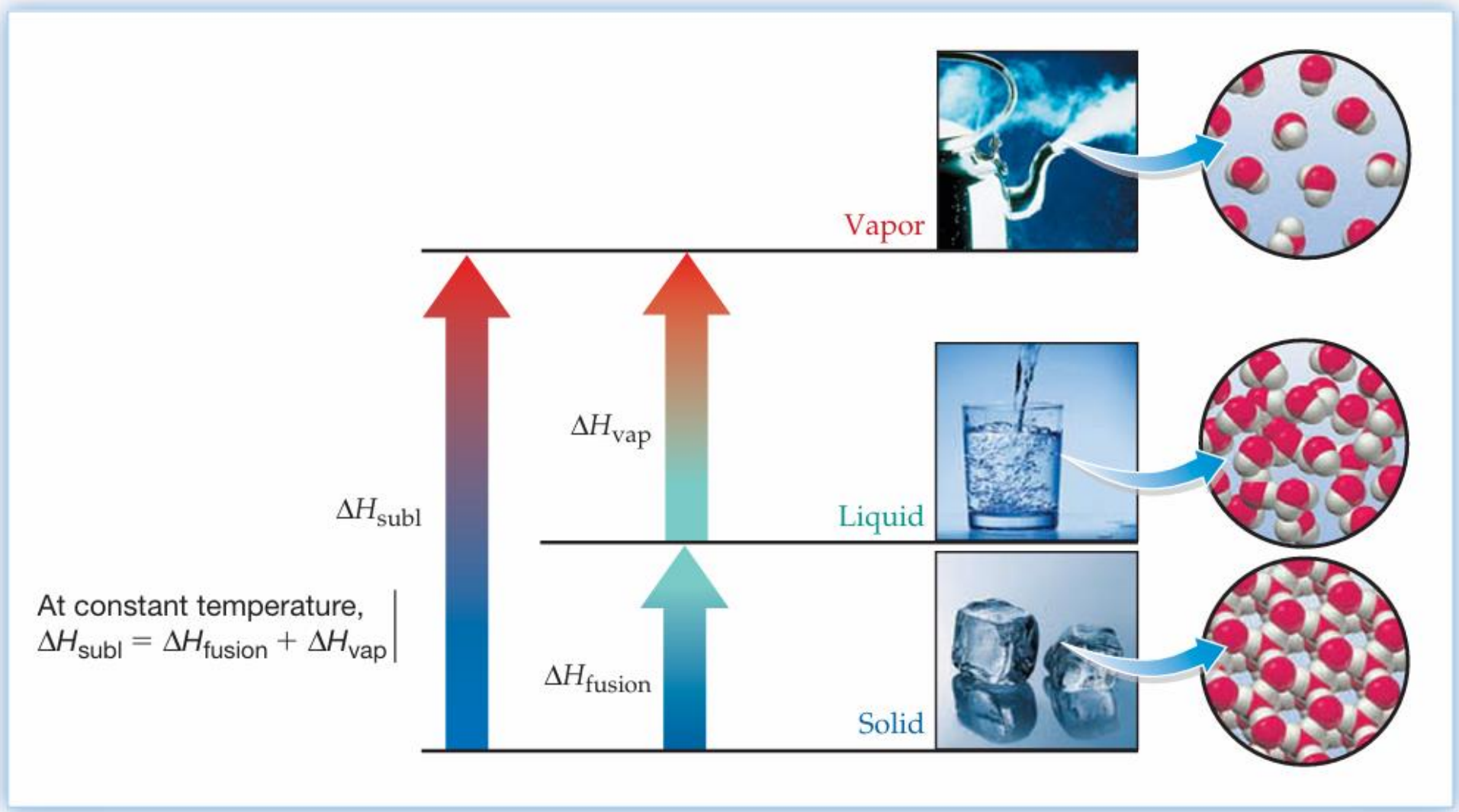


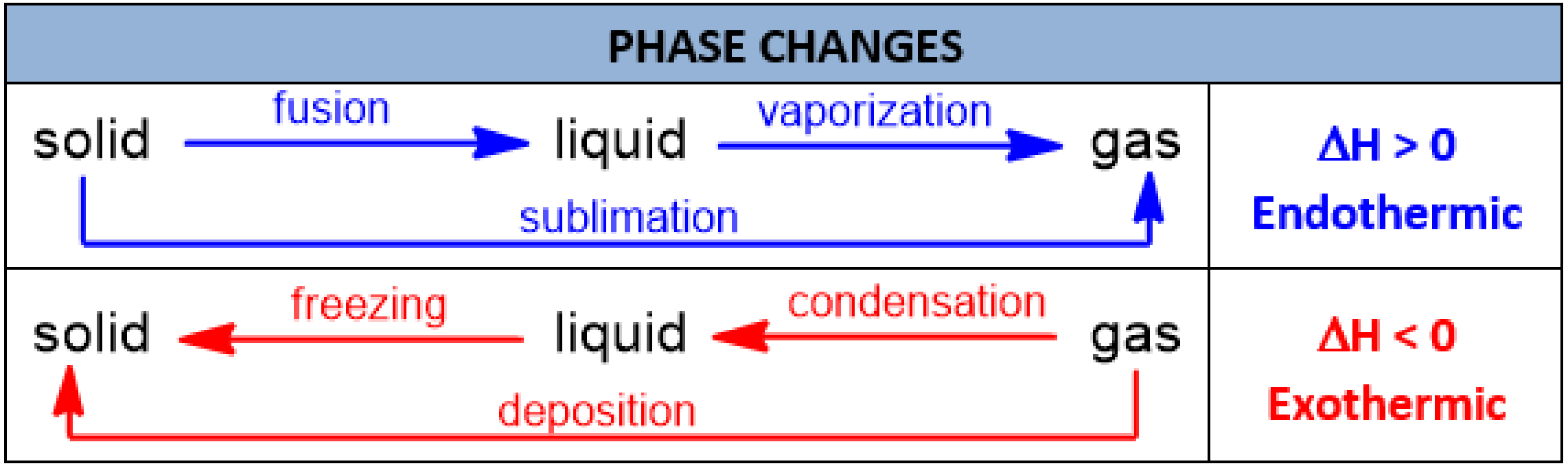
$$\Delta H = H_{\text{final}} - H_{\text{initial}}$$
$$= H_{\text{products}} - H_{\text{reactants}}$$

Enthalpy is the sum of a thermodynamic system's internal energy and the product of its pressure and volume.

Standard enthalpy(ΔH°)is the enthalpy change measured under **standard thermodynamic conditions (1 atm, 25 °C, and 1 M** concentration for all substances in solution).

1.5. Physical change enthalpy





- Endothermic (endo means “within", so heat flows in)
  ΔH has a positive sign.
- Exothermic (exo means “out” ,so heat flows out)
  ΔH has a negative sign

1.6. Chemical change enthalpy [Standard heats of formation]

$(\Delta H°_f)$ is the enthalpy change due to the formation of 1 mol of a substance in its standard state.

$$\Delta H°_{\text{reaction}} = \Delta H°_f \text{ (Products)} - \Delta H°_f \text{ (Reactants)}$$

To find $\Delta H°$ for the reaction

$$a\,\text{A} + b\,\text{B} + \cdots \longrightarrow c\,\text{C} + d\,\text{D} + \cdots$$

Subtract the sum of the heats of formation for these reactants . . .

. . . from the sum of the heats of formation for these products.

$$\Delta H°_{\text{reaction}} = [c\,\Delta H°_f(\text{C}) + d\,\Delta H°_f(\text{D}) + \cdots] - [a\,\Delta H°_f(\text{A}) + b\,\Delta H°_f(\text{B}) + \cdots]$$

Note: the standard heat of formation of an element (O_2, N_2,H_2, C,..etc) in its standard state is equal to zero.

Solved example

Calculate ΔH° in kilojoules for the synthesis of lime (CaO) from limestone ($CaCO_3$), the key step in the manufacture of cement.

$$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$$

$$\Delta H^\circ_f\,[CaCO_3(s)] = -1207.6\ \text{kJ/mol}$$
$$\Delta H^\circ_f\,[CaO(s)] = -634.9\ \text{kJ/mol}$$
$$\Delta H^\circ_f\,[CO_2(g)] = -393.5\ \text{kJ/mol}$$

Solution

$$\Delta H^\circ = [\Delta H^\circ_f\,(CaO) + \Delta H^\circ_f\,(CO_2)] - [\Delta H^\circ_f\,(CaCO_3)]$$
$$= (1\ \text{mol})(-634.9\ \text{kJ/mol}) + (1\ \text{mol})(-393.5\ \text{kJ/mol}) - (1\ \text{mol})(-1207.6\ \text{kJ/mol})$$
$$= 179.2\ \text{kJ}$$

The reaction is endothermic by 179.2 kJ.